

Project Report

I.Title: Disaster-Resilient Communication System

II. Nature of work:

Novel solution/ indigenous design to a problem/ challenge

III. Rationale: During Natural Disaster, power and mobile networks collapse, cutting off communication when it's most critical. Existing systems depend on the grid and expensive infrastructure. Our project aims to address this gap by creating low-cost, off-grid, and infrastructure-retrofitting systems. We leverage the widespread and firm infrastructure of street lamps, which are already being retrofitted with solar lamps, we propose to add another level to this, which would enable communication in a Disaster.

IV.Scientific Principle(s)/ Concepts:

Light	Electricity	Hashing
Light fidelity	Circuits	
Line of Sight	Ohm's Law	
Propagation of light	Embedded Systems	

V.Materials Used:

Cardboard
Plyboard
Steel straws
Chart paper

- NodeMCU (ESP8266)
- IR transmitter Leds
- IR TSOP(1738) Receivers
- Normal LEDs
- Resistors
- Transistors
- Wires
- Batteries
- Buttons
- Solar Panel

Instruments / Tools :

Laptop
Cable
Powerbank

VI.Procedure/ Description:

Part 1: Hardware Model

Components(Per Node)

IR Transmitter

We connected 2-3 IR LEDs (940 nm wavelength) in series with a 100Ω resistor to send signals to a NPN transmitter(2N2222). The transistor base connects to GPIO pin through a 2.2kΩ resistor.

IR Receiver(TSOP38238)

filters ambient light and demodulates 38 kHz carrier signals

SOS Button(For lamp nodes only)

push button connects between GPIO(internal pull-up resistor) pin and ground

Power Supply

For lamp nodes we are using the existing solar powered battery system whereas router and hq nodes depend on external power supply
In our demo power is supplied using USB or 3.7V sources.

Part 2 : System Design

Part 2.1 High-Level System Design

This system defines three modular node roles that coordinate to form a disaster-resilient optical mesh:

HQ Node

Acts as the central controller.
Connected to computer for monitoring dashboard
Receives all SOS alerts and status messages from network
Sends downlink broadcasts (e.g., alerts, responses) back into the mesh.
Maintains a log of all incoming messages via a serial-connected dashboard.
Initiates the mesh by sending INIT packets that set the hop count gradient.

Lamp Nodes

Generate SOS alerts when button pressed
Forwards received messages
Broadcasts messages to civilians via visible light
Solar-powered (off-grid operation)
Repeat the most recent broadcast periodically to alert late arrivals.

Router Node

Extends network range
Provides redundant communication paths
Strengthens mesh connectivity
Allows big societies to have coverage and support with them
Used for text based Communication

Two-Way Communication System:

The mesh supports bidirectional message movement

Uplink (Lamp → HQ):

- SOS alerts propagate from lamps toward HQ using hop gradient
- Each node forwards toward lower hop count (closer to HQ)

Downlink (HQ → Lamps):

- HQ broadcasts emergency instructions to all lamps
- HQ sends targeted messages to specific lamp IDs
- Lamps retransmit to neighbors (flood-fill propagation)

To guide message forwarding intelligently, the HQ sends an INIT packet[type 0 message] containing hop = 0. Lamp nodes record their hop distance and only forward messages if it brings the packet closer to HQ, with a small relaxation ($K = 1$ or 2) to allow rerouting when needed.

Message(Lamp to HQ) Routing:

When Lamp1 (hop=5) sends SOS:

- > Includes current hop count in header
- > Lamp2 (hop=4) receives, decrements: hop becomes 4
- > Only forwards if its own hop $\leq$ message hop + tolerance
- > Prevents messages from moving "away" from HQ
- > Message flows downhill toward HQ (decreasing hop counts)

Node Processing Functions:

Deduplication:

- Each node maintains circular cache of last n message hashes and source
- Before forwarding, computes message hash
- Checks cache: if hash exists → discard, if new → forward
- Prevents infinite loops and broadcast storms

IR Retransmission:

- Messages sent multiple times using FIFO
- Ensures delivery even if first attempt fails (collision/interference)
- Automatic redundancy without acknowledgments

Repeating Last Broadcast:

- Lamp nodes remember last HQ broadcast message
- Retransmit to visible light (for phones) every 60 seconds
- Ensures people entering area receive alerts even after initial broadcast

Part 2.2 Header and Packet System

Two-Burst Transmission:

- Burst 1: Header
- Brief gap (50 ms)
- Burst 2: Message content (variable length)

- Receiver waits for both bursts before processing
- n secs timeout also defined for non-blocking lost messages

Header Format System

Ideal Header:
[000h][FFFF][1][3A7F][09]
↑ ↑ ↑ ↑ ↑ ↑
Src Dst Type Hash msgHop

Length: 13-character header + message

Message Flow Example:

Scenario: Lamp4(102a) sends SOS

1. Button pressed → generates header `102a000h3`
2. Transmits in all 4 directions (IR LEDs)
3. Lamp3 receives via TSOP
5. Checks cache: not seen before
6. Adds to cache, forwards header in all 4 directions
7. Other nodes receives, repeats process
8. After 4 hops, HQ receives and logs: "SOS from 102a"

Part 2.3 Issues And Solutions

Problem	Engineering Challenge	Our Solution
Loop Prevention	Mesh networks create infinite message circulation	Cache-based deduplication + gradient routing
Reliable Delivery	No ACKs in disaster scenarios (latency-critical)	3× redundant transmission with timed intervals
Message Integrity	IR noise causes corruption	16-bit polynomial hash for error detection and timeout
Power Constraints	Limited battery capacity, must run 24/7	Event-driven TX (100mA bursts), idle <0.06W
Protocol Complexity	4 message types, varying lengths (9-15 chars)	Fixed-length headers with type-based parsing

3. Variables, Parameters & Testing

Independent Variables:

- * Distance between each node (kept ~10 cm for prototype)

Controlled Variables:

- * Environment: Indoor room with ambient light
- * Transmission circuit: IR LEDs in series, 100Ω resistor, driven by 2N2222
- * Power source: Constant 3.7V supply from USB or battery

Measured / Dependent Variables:

- * Time taken per hop: ~1-2 seconds for a 13-15 character header
- * Packet size: Header typically 9-15 characters; payload variable

Procedure:

Hardware

1. Placed ESP8266 NodeMCU on breadboard center
2. Connected IR transmitter circuit:
 - Wired D1 pin to 2N2222 transistor base
 - Connected IR LED series (2-3 LEDs) from 5V through 100/220Ω resistor to transistor collector
 - Connected transistor emitter to Ground
3. Connected TSOP receiver:
 - VCC to 3.3V, GND to Ground, OUT to D2
4. Added push button between D3 and Ground
5. Connected status LED through 220Ω resistor from D4 to Ground

Firmware

1. Installed Arduino IDE and ESP8266 support
2. Installed IRremote library for IR communication
3. Configure config.h to set variables and pins
3. Upload the code for the specific node

Full Code can be Seen at: <https://github.com/vassu-v/D-LiFi-Proto>

Our Hypothesis:

Primary Prediction:

If we press SOS on Lamp1, the message will propagate hop-by-hop through the network and reach HQ within 10 seconds, with each intermediate node forwarding exactly once despite multiple receive paths.

Expected Behavior:

- > Single hop delay: 1-2 seconds
- > Total 5-hop chain: 5-10 seconds

- > Zero duplicate forwards (cache prevents loops)
- > Network continues functioning if one node removed (in mesh topology)

4. Observations & Data Highlights

- * Range: Stable transmission observed up to 20 cm with simple IR LED and 100Ω resistor
- * Latency: ~1–2 seconds per hop in clean conditions (indoors)
- * Message propagation works reliably for up to 5–6 hops
- * Deduplication is effective — messages are not endlessly rebroadcast
- * No significant interference from ambient room lighting during testing

Implications:

The system:

- * Works in indoor and low-light conditions
- * Maintains connectivity despite node failure (self-healing)
- * Leverages affordable, replicable components
- * Can scale with additional nodes or media (e.g., LoRa or ultrasound)
- * With directional hardware and topology discovery, this design can scale to real-world deployment using retrofitted solar street lamps
- * Cities can integrate this communication layer into Smart City infrastructure as a low-cost, backup alert system for disasters
- * The modular software architecture supports hybrid modes, future encryption, and advanced routing protocols

VII. References

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